Exponent

03



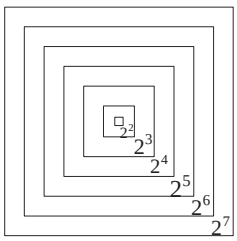
Nidhi, Mayank and Reshmi were asking each other questions about numbers-

Nidhi- What will we get when we multiply ten thousand by one lakh?

Reshmi- Hundred million which is the same as one billion. But do you know how many zeroes are there in this number?

Mayank- These will be 9 zeroes because ten thousand is 10^4 and one lakh is 10^5 and $10^4 \times 10^5 = 10^9$.

Mayank- My turn now. There are seven boxes. Each box is inside a bigger one. The smallest one has 2 beads and the next bigger box has twice as many and so on for every next box. Then how many beads are there in the seventh box? It is very difficult, it will take you a long time.



Nidhi- Why? 7 box and the numbers is doubled each time that means $2 \times 2 \times 2 \times 2 \times 2 \times 2 \times 2$ means $2^7 = 128$.

Reshmi- But let us try and also find out the total numbers of beads.

Discuss among yourselves and find out the total number of beads.

Try This



1. There are four rice grains in one of the square of the chess board. In the second square next to it, the number of grains is four times of that, in the third square number is four times of second and so on. Tell me how many grains of rice will be there in the fourth square. Write it in exponent form?
2. Solve these-

   (i) $3^5 \times 3^7$ (ii) $17 \times 17 \times 17 \times 17$ (iii) $5 \times 3 \times 3 \times 3 \times 3$

   (iv) $\left(\frac{2}{3}\right)^3 \times \left(\frac{2}{3}\right)^5$ (v) $\frac{2}{5} \times \frac{2}{5} \times \frac{2}{5}$
3. Distance between Sun-Earth is approximately 150000000 kilometer. Write it as an exponent?

   You can also prepare three question of this kind and give friends to attempt to solve them.

Laws of Exponent

If a is a real number and m, n are integers then,

1. Rule of multiplication is $a^m \times a^n = a^{m+n}$

2. Rule of division is $\frac{a^m}{a^n} = a^{m-n} (a \neq 0)$

3. Rule of powers of exponent is $\left(a^m\right)^n = a^{mn}$

4. $(a\,b)^m = a^m \times b^m$

5. $\left(\frac{a}{b}\right)^m = \frac{a^m}{b^m}$

6. Meaning of a^0



$\frac{a^m}{a^n} = a^{m-n}$ where $a \neq 0$

if $m = n$, then

$\frac{a^m}{a^m} = a^{m-m}$

$1 = a^0$

Therefore $a^0 = 1$

7. $a^m = \frac{1}{a^{-m}}$; or $a^{-m} = \frac{1}{a^m}$

EXAMPLE-1. Simplify these-

(i) $3^5 \times 9^4$ (ii) $(4 \times 3)^4 \times \left(\frac{2}{4}\right)^6$ (iii) $\frac{(6^2)^4 \times 6^5}{6^4}$

SOLUTION : (i) $3^5 \times 9^4$

$= 3^5 \times (3^2)^4 = 3^5 \times 3^{2\times4}$ $\left[(a^m)^n = a^{m\times n}\right]$

$= 3^5 \times 3^8 = 3^{(5+8)}$ $\left[a^m \times a^n = a^{m+n}\right]$

$= 3^{13}$

(ii) $(4\times 3)^4 \times \left(\frac{2}{4}\right)^6$

$= \left(4^4 \times 3^4\right) \times \frac{2^6}{4^6} \qquad \because \left[\left(\frac{a}{b}\right)^m = \frac{a^m}{b^m}\right]$ or $\left[(ab)^m = a^m \times b^m\right]$

$= \left((2^2)^4 \times 3^4\right) \times \frac{2^6}{(2^2)^6}$

$= (2^{2\times 4} \times 3^4) \times \frac{2^6}{2^{2\times 6}} \qquad \left[(a^m)^n = a^{m\times n}\right]$

$= 2^8 \times 3^4 \times \frac{2^6}{2^{12}} = 2^8 \times 3^4 \times 2^{6-12} \qquad \left[\frac{a^m}{a^n} = a^{m-n}\right]$

$= 2^8 \times 3^4 \times 2^{-6}$

$= 2^{8-6} \times 3^4 = 2^2 \times 3^4 \qquad \left[a^m \times a^n = a^{m+n}\right]$

(iii) $\frac{\left(6^2\right)^4 \times 6^5}{6^4} = \frac{6^{2\times 4} \times 6^5}{6^4} \qquad \left[(a^m)^n = a^{m\times n}\right]$

$= \frac{6^8 \times 6^5}{6^4} = \frac{6^{8+5}}{6^4} = \frac{6^{13}}{6^4} \qquad \left[a^m \times a^n = a^{m+n}\right]$

$= 6^{13-4} = 6^9 \qquad \left[\frac{a^m}{a^n} = a^{m-n}\right]$

Try This

Simplify the following-

(i) $\frac{5^4 \times 5^6}{5^3}$ (ii) $\frac{(2^2)^3 \times 8^7}{4^4}$

(iii) $\frac{(9\times 3)^8}{(3)^5}$ (iv) $(3\times 2)^4 \times \left(\frac{2}{3}\right)^6$



Negative Exponents

We know that numbers can be written in the form of other numbers and their exponents-

Like-

1 kilometer	= 1000 meter	$= 10^3$ meter
1 hectometer	= 100 meter	$= 10^2$ meter
1 decameter	= 10 meter	$= 10^1$ meter
1 meter	= 1 meter	= ?

Here the number are expressed in the standard form of powers of 10.

If we have a number smaller than 1, then how do we write that -

To do that let us look at the following pattern-



1 decimeter $= \frac{1}{10}$ meter $= 10^{-1}$ meter

1 centimeter $= \frac{1}{100}$ meter $= 10^{-2}$ meter

1 milimeter $= \frac{1}{1000}$ meter $= 10^{-3}$ meter

As shown above, we can write $\frac{1}{10} = \frac{1}{10^1} = 10^{-1}$

$$\frac{1}{100} = \frac{1}{10 \times 10} = \frac{1}{10^2} = 10^{-2}$$

$$\frac{1}{1000} = \frac{1}{10 \times 10 \times 10} = \frac{1}{10^3} = 10^{-3}$$

This means $\frac{1}{10^n} = 10^{-n}$

Then is $\frac{1}{10^{-n}} = 10^n$? Discuss.



We write 4^{-3} as $\frac{1}{4^3}$. Similarly, we can write-

$$5^{-6} = \frac{1}{5^6}$$

Let us consider some more examples-

$$\frac{1}{6^{-3}} = 6^3$$

$$\frac{1}{9^2} = 9^{-2}$$

$$3^3 = \frac{1}{3^{-3}}$$

From the above examples we can see-

$1 = 6^3 \times 6^{-3}$

$1 = 9^2 \times 9^{-2}$

$3^3 \times 3^{-3} = 1$

Using these we can say that for any rational number 'a' (other than '0') $a^{-m} = \frac{1}{a^m}$, which is the multiplicative inverse of a^m.

because $a^m \times a^{-m} = a^{m+(-m)} = a^0 = 1$, where 'm' is an integer.

Try This



1. Express the following numbers in exponent form-

   (i) $\frac{1}{8}$ (ii) $\frac{1}{243}$ (iii) $\frac{1}{196}$

2. Write the multiplicative inverse of-

   (i) 10^{-5} (ii) $\frac{1}{2^3}$ (iii) P^{-n} (iv) 5^{-7}

3. Simplify each of the following-

   (i) $\left((5^2)^3 \times 5^4\right) \div 5^6$ (ii) $2^2 \times \frac{3^2}{2^{-2}} \times 3^{-1}$ (iii) $(14^{-2} \times 13^{-2}) \div 6^{-1}$

EXAMPLE-2. Simplify each of the following-

(i) $3^4 \times 3^{-8}$ (ii) $(-2)^{-3} \times (-2)^{-4}$

SOLUTION : (i) We know that- $3^{-8} = \frac{1}{3^8}$ $\left[a^{-m} = \frac{1}{a^m}\right]$

Hence, $3^4 \times 3^{-8} = 3^4 \times \frac{1}{3^8} = \frac{3^4}{3^8}$

$= 3^{4-8} = 3^{-4}$

(ii) $(-2)^{-3} \times (-2)^{-4} = \frac{1}{(-2)^3} \times \frac{1}{(-2)^4} = \frac{1}{(-2)^{3+4}}$

$= \frac{1}{(-2)^7} = (-2)^{-7}$

EXAMPLE-3. Find the values of -

(i) 5^{-2} (ii) $\frac{1}{2^{-5}}$ (iii) $\frac{4^7}{4^4}$

SOLUTION : (i) $5^{-2} = \frac{1}{(5)^2} = \frac{1}{5 \times 5} = \frac{1}{25}$

(ii) $\frac{1}{2^{-5}} = 2^5 = 2 \times 2 \times 2 \times 2 \times 2 = 32$

(iii) $\frac{4^7}{4^4} = 4^{7-4} = 4^3 = 4 \times 4 \times 4 = 64$

EXAMPLE-4. Find the values of -

(i) $\left(\frac{4}{7}\right)^{-3}$ (ii) $4^4 \times 16^{-2} \times 4^0$

SOLUTION : (i) $\left(\frac{4}{7}\right)^{-3} = \frac{(4)^{-3}}{(7)^{-3}} = 4^{-3} \times \frac{1}{7^{-3}}$ $\left[\left(\frac{a}{b}\right)^m = \frac{a^m}{b^m}\right]$

$$= \frac{7^3}{4^3} \quad \left[a^{-m} = \frac{1}{a^m} \text{ or } a^m = \frac{1}{a^{-m}}\right]$$

$$= \frac{7 \times 7 \times 7}{4 \times 4 \times 4} = \frac{343}{64}$$



(ii) $4^4 \times 16^{-2} \times 4^0$

$$= \frac{4^4 \times 4^0}{16^2} \qquad \left[a^{-m} = \frac{1}{a^m}\right]$$

$$= \frac{4^4 \times 4^0}{\left(4^2\right)^2}$$

$$= \frac{4^4 \times 4^0}{4^4} \qquad \left[\left(a^m\right)^n = a^{m \times n}\right]$$

$= 4^{4+0-4} = 4^0$

$= 1$

EXAMPLE-5. Simplify- $\left[\left\{\left(\frac{1}{3}\right)^{-3} - \left(\frac{1}{2}\right)^{-3}\right\} \div \left(\frac{1}{5}\right)^{-2}\right]$

SOLUTION : $\left[\left\{\left(\frac{1}{3}\right)^{-3} - \left(\frac{1}{2}\right)^{-3}\right\} \div \left(\frac{1}{5}\right)^{-2}\right]$ $\left[\text{we know that } \left(\frac{a}{b}\right)^m = \frac{a^m}{b^m}\right]$

$$= \left[\left(\frac{1^{-3}}{3^{-3}} - \frac{1^{-3}}{2^{-3}}\right) \div \frac{1^{-2}}{5^{-2}}\right] \qquad \left[\text{we know that } a^{-m} = \frac{1}{a^m} \text{ and } a^m = \frac{1}{a^{-m}}\right]$$

$$= \left[\left(\frac{3^3}{1^3} - \frac{2^3}{1^3}\right) \div \frac{5^2}{1^2}\right] = \left(\frac{27}{1} - \frac{8}{1}\right) \div 25$$

$$= (27 - 8) \div 25 = \frac{19}{25}$$

EXAMPLE-6. Simplify- $\left(\frac{2}{5}\right)^{-3} \times \left(\frac{25}{4}\right)^{-2}$

SOLUTION : $\left(\frac{2}{5}\right)^{-3} \times \left(\frac{25}{4}\right)^{-2}$

$= \left(\frac{2}{5}\right)^{-3} \times \left(\frac{5^2}{2^2}\right)^{-2} \quad \left\{\because \frac{25}{4} = \frac{5\times5}{2\times2} = \frac{5^2}{2^2}\right\} \quad \left[\because (a^m)^n = a^{mn}\right]$

$= \frac{5^3}{2^3} \times \frac{2^4}{5^4} = 5^{3-4} \times 2^{4-3} \qquad \left[like \frac{1}{a^m} = a^{-m} \; and \; \frac{1}{a^{-m}} = a^m\right]$

$= 5^{-1} \times 2^1 = \frac{2}{5}$

EXAMPLE-7. If $x = \left(\frac{3}{2}\right)^2 \times \left(\frac{2}{3}\right)^{-4}$ then find the value of x^{-2}.

SOLUTION : $x = \left(\frac{3}{2}\right)^2 \times \left(\frac{2}{3}\right)^{-4}$

$x = \frac{3^2}{2^2} \times \frac{2^{-4}}{3^{-4}} \quad \left[we\,know \left(\frac{a}{b}\right)^m = \frac{a^m}{b^m}\right]$

$x = \frac{3^2}{2^2} \times \frac{3^4}{2^4} = \frac{3^{2+4}}{2^{2+4}} = \frac{3^6}{2^6} = \left(\frac{3}{2}\right)^6$

$x^{-2} = \left[\left(\frac{3}{2}\right)^6\right]^{-2} = \left(\frac{3}{2}\right)^{-12}$

$x^{-2} = \frac{3^{-12}}{2^{-12}} = \frac{2^{12}}{3^{12}} = \left(\frac{2}{3}\right)^{12}$



EXAMPLE-8. Simplify-

(i) $\frac{3^{2m+1} \times 9^{3m}}{3^{4m+1}}$ (ii) $\frac{x^{a+b}y^{a-b}}{x^{a+2b}y^{a-2b}}$



SOLUTION : (i) $\dfrac{3^{2m+1} \times 9^{3m}}{3^{4m+1}}$

$= \dfrac{3^{2m+1} \times (3^2)^{3m}}{3^{4m+1}}$

$= \dfrac{3^{2m+1} \times 3^{6m}}{3^{4m+1}} = \dfrac{3^{2m+1+6m}}{3^{4m+1}}$

$= \dfrac{3^{8m+1}}{3^{4m+1}} = 3^{8m+1-4m-1}$

$= 3^{4m}$

(ii) $\dfrac{x^{a+b}y^{a-b}}{x^{a+2b}y^{a-2b}}$

$= x^{a+b} \times y^{a-b} \times \overline{x}^{(a+2b)} \times \overline{y}^{(a-2b)}$

$= x^{a+b-a-2b} \times y^{a-b-a+2b}$

$= x^{-b} \times y^{b}$

$= \dfrac{1}{x^b} \times y^b = \left(\dfrac{y}{x}\right)^b$

Expanded Form of Decimal Numbers

Expanded form of 328 is-

$328 = 300 + 20 + 8$

$= 3 \times 100 + 2 \times 10 + 8 \times 1$

$= 3 \times 10^2 + 2 \times 10^1 + 8 \times 10^0$

Similarly, $4158 = 4 \times 1000 + 1 \times 100 + 5 \times 10 + 8 \times 1$

$= 4 \times 10^3 + 1 \times 10^2 + 5 \times 10^1 + 8 \times 10^0$



Similarly, if we expand 132.28 we get the number in the form of exponents-

$$132.28 = 1 \times 100 + 3 \times 10 + 2 \times 10^0 + 2 \times \frac{1}{10} + 8 \times \frac{1}{100}$$

$$= 1 \times 10^2 + 3 \times 10^1 + 2 \times 10^0 + 2 \times 10^{-1} + 8 \times 10^{-2}$$

Try This



Write the following numbers in expanded form of exponents-

(i)	**15.1**	**(ii)**	**512.23**
(iii)	**537.204**	**(iv)**	**205.003**

Standard Representations of Very Large and Very Small Numbers

Large number such as- The diameter of the Sun is approximately 1400000000 meters, can be written in the standard form as 1.4×10^9 meter. The number will be written as powers of 10. Similarly, 6.2×10^5 meter will be expressed in the common form as- 6.2×100000 meter = 620000 meter.

Similarly, a very small number like- The charge of an electron is 0.00000000000000000016 coulomb, can be written as 1.6×10^{-19} coulomb.

Here we write numbers in the form of powers of 10 and represent both large and small numbers, in the standard form.

Comparison between Very Large and Very Small Numbers

The distance between the Sun and the Earth is 1.496×10^{11}m and the distance between the Earth and the Moon in 3.703×10^8m. During a solar eclipse, the Moon comes between the Sun and the Earth.What will be the distance between the Moon and Sun at this time?

$= 1.496 \times 10^{11} - 3.703 \times 10^8$

$= 1.496 \times 1000 \times 10^8 - 3.703 \times 10^8$

$= (1496 - 3.703) \times 10^8$

$= 1492.297 \times 10^8$m

EXAMPLE-9. Express the following in standard form -

(i) 40600000000 (ii) 2150000000000

SOLUTION : (i) 4.06×10^{10}

(ii) 2.15×10^{12}

EXAMPLE-10. Express the following in the decimal form-

(i) 3×10^{-8} (ii) 4.37×10^{-5}

SOLUTION : (i) 3×10^{-8}

$$=\frac{3}{10^8}=\frac{3}{100000000}=0.00000003$$

(ii) 4.37×10^{-5}

$$=\frac{4.37}{10^5}=\frac{4.37}{100000}=0.0000437$$

Exercise - 3.1



1. Find the values of -

(i) $\left(\frac{1}{2}\right)^{-5}$ (ii) $\frac{1}{3^{-4}}$ (iii) $\frac{6^7}{2^3\times3^7}$

2. Simplify each of the following-

(i) $(-4)^3\times(-2)^{-3}$ (ii) $(-3)^4\times\left(\frac{5}{3}\right)^4$ (iii) $(-5)^3\div(5)^{-7}$

3. Simplify-

(i) $\frac{16\times t^{-3}}{4^{-3}\times8\times t^{-6}}$ $(t\neq0)$ (ii) $\left\{\left(\frac{1}{3}\right)^{-1}-\left(\frac{1}{4}\right)^{-1}\right\}^{-1}$



4. Prove-

(i) $\left(\frac{x^a}{x^b}\right)^{a+b}\times\left(\frac{x^b}{x^c}\right)^{b+c}\times\left(\frac{x^c}{x^a}\right)^{c+a}=1$

(ii) $\frac{1}{1+x^{m-n}}+\frac{1}{1+x^{n-m}}=1$

5. Write the following numbers in the standard form-

(i) 0.00000000000852 (ii) 8020000000000000

(iii) 41960000000



6. Express the following numbers in the decimal form-

(i) 5.02×10^{-6} (ii) 7×10^{-8}

(iii) 1.00001×10^{9}

7. Write numbers in the following statements in standard form-

(i) Size of red blood cells is 0.000007 m.

(ii) The diameter of the Earth is 12756000 m.

(iii) Thickness of a paper is 0.08 m.

Positive rational Exponent

We know that $2^3 = 8$

We may express this relation as $8^{\frac{1}{3}} = 2$

Similarly, $5^3 = 125$ can also be expressed as $(125)^{\frac{1}{3}} = 5$.

In general, if x and y are non-zero rational numbers and m is a positive integer such that $x^m = y$, then we may write $y^{\frac{1}{m}} = x$. We may also write $y^{\frac{1}{m}}$ as $\sqrt[m]{y}$ and call it the mth root of y.

For example the second root of $9 = \sqrt{9} = \sqrt[2]{9} = 3$

Some other example are, the volume of a cube is 64 units so its side will be $64^{\frac{1}{3}}$ units means-

Third root of 64 = $\sqrt[3]{64} = 4$ units

That is the side of cube will be 4 units.

Fourth root of 625 = $\sqrt[4]{625} = 5$

The second root is called 'square root' and the third root is the 'cube root'.

Thus we can define x^m as a positive rational exponent m.

If x is a positive rational number and $m = \frac{p}{q}$ is a positive rational exponent, then we define $x^{\frac{p}{q}}$ as the qth root of x^p.

For example the volume of a sphere is $\frac{4}{3}\pi \times (125)$ implies $r^3 = 125$

So its radius will be-

$$r = \sqrt[3]{125} = 125^{\frac{1}{3}} = (5^3)^{\frac{1}{3}}$$

Radius $r = 5^{3/3} = 5^1 = 5$

That is, $x^{\frac{p}{q}} = \left(x^p\right)^{\frac{1}{q}}$

For example, $(8)^{\frac{5}{3}}$ can be expressed in various ways-

$(8)^{\frac{5}{3}}$ or $\left(8^5\right)^{\frac{1}{3}}$ or $\left(8^{\frac{1}{3}}\right)^5$ or $\left(2^{3\times\frac{1}{3}}\right)^5$

or 2^5 or 32

If x is a positive rational number then for positive rational exponent $\frac{p}{q}$,

$$\left(x^p\right)^{\frac{1}{q}} = \left(x^{\frac{1}{q}}\right)^p$$

EXAMPLE-11. Find the value of-

(i) $(27)^{\frac{2}{3}}$ (ii) $\left(\frac{32}{243}\right)^{\frac{4}{5}}$



SOLUTION : (i) $(27)^{\frac{2}{3}} = \left(27^2\right)^{\frac{1}{3}} = (729)^{\frac{1}{3}}$

$= (3\times3\times3\times3\times3\times3)^{\frac{1}{3}}$

$= 3^{6\times\frac{1}{3}} = 3^2 = 9$

or $(27)^{\frac{2}{3}} = \left(27^{\frac{1}{3}}\right)^2$

$= \left(3^{3\times\frac{1}{3}}\right)^2 = (3)^2 = 9$

(ii) $\left(\frac{32}{243}\right)^{\frac{4}{5}} = \left\{\left(\frac{2\times2\times2\times2\times2}{3\times3\times3\times3\times3}\right)^{\frac{1}{5}}\right\}^4$

$= \left\{\left(\frac{2^5}{3^5}\right)^{\frac{1}{5}}\right\}^4 = \left\{\left(\frac{2}{3}\right)^{5\times\frac{1}{5}}\right\}^4 = \left(\frac{2}{3}\right)^4$

$= \frac{2\times2\times2\times2}{3\times3\times3\times3} = \frac{16}{81}$

or $\left(\frac{32}{243}\right)^{\frac{4}{5}} = \left[\left(\frac{32}{243}\right)^4\right]^{\frac{1}{5}}$



$= \left[\left(\frac{2^5}{3^5}\right)^4\right]^{\frac{1}{5}} = \left[\frac{2^{5\times4}}{3^{5\times4}}\right]^{\frac{1}{5}}$

$= \left(\frac{2^{20}}{3^{20}}\right)^{\frac{1}{5}} = \frac{2^4}{3^4}$

$= \left(\frac{2}{3}\right)^4 = \frac{16}{81}$

In the above example, we have used both the forms. Which form do you think is easier for the purpose of calculation?

The exponent rules also apply to rational exponents. Let us see.

The first method $\left(\frac{4}{25}\right)^{\frac{1}{2}} \times \left(\frac{4}{25}\right)^{\frac{3}{2}}$



$$\left\{\left(\frac{2}{5}\right)^{2}\right\}^{\frac{1}{2}} \times \left\{\left(\frac{2}{5}\right)^{2}\right\}^{\frac{3}{2}}$$

or $$\left\{\frac{2}{5}\right\}^{2\times\frac{1}{2}} \times \left(\frac{2}{5}\right)^{\cancel{2}\times\frac{3}{\cancel{2}}}$$

$$\left(\frac{2}{5}\right)^{1} \times \left(\frac{2}{5}\right)^{3} = \frac{2}{5}\times\frac{2}{5}\times\frac{2}{5}\times\frac{2}{5} = \frac{16}{625}$$

The second method $$\left(\frac{4}{25}\right)^{\frac{1}{2}} \times \left(\frac{4}{25}\right)^{\frac{3}{2}} = \left(\frac{4}{25}\right)^{\frac{1}{2}+\frac{3}{2}}$$

$$\left(\frac{4}{25}\right)^{\frac{1+3}{2}} = \left(\frac{4}{25}\right)^{\frac{\cancel{4}^{2}}{\cancel{2}}} = \left(\frac{4}{25}\right)^{2}$$

$$= \frac{4\times4}{25\times25} = \frac{16}{625}$$

We get the same value by both the methods. Hence $\left(\frac{4}{25}\right)^{\frac{1}{2}} \times \left(\frac{4}{25}\right)^{\frac{3}{2}} = \left(\frac{4}{25}\right)^{\frac{1}{2}+\frac{3}{2}}$.

Rational exponent also follow the rules of exponents $x^{m} \times x^{n} = x^{m+n}$.

Is $\left(\frac{16}{81}\right)^{\frac{5}{4}} \div \left(\frac{16}{81}\right)^{\frac{3}{4}} = \left(\frac{16}{81}\right)^{\frac{5}{4}-\frac{3}{4}}$, let's check this-

$$\left(\frac{16}{81}\right)^{\frac{5}{4}} \div \left(\frac{16}{81}\right)^{\frac{3}{4}} = \left(\frac{16}{81}\right)^{\frac{5}{4}-\frac{3}{4}}$$

$$\left[\left(\frac{2}{3}\right)^{4}\right]^{\frac{5}{4}} \div \left[\left(\frac{2}{3}\right)^{4}\right]^{\frac{3}{4}} = \left[\left(\frac{2}{3}\right)^{4}\right]^{\frac{5}{4}-\frac{3}{4}}$$

$$\left(\frac{2}{3}\right)^{\cancel{4}\times\frac{5}{\cancel{4}}} \div \left(\frac{2}{3}\right)^{\cancel{4}\times\frac{3}{\cancel{4}}} = \left(\frac{2}{3}\right)^{\cancel{4}\times\frac{2}{\cancel{4}}}$$

$$\left(\frac{2^{5}}{3^{5}}\right) \div \left(\frac{2^{3}}{3^{3}}\right) = \left(\frac{2}{3}\right)^{2}$$



$$\left(\frac{2^5}{3^5}\times\frac{3^3}{2^3}\right)=\left(\frac{2^2}{3^2}\right)$$

$$\left(\frac{2^{5-3}}{3^{5-3}}\right)=\left(\frac{2^2}{3^2}\right)$$

$$\left(\frac{2^2}{3^2}\right)=\left(\frac{2^2}{3^2}\right) \qquad \text{R.H.S.} = \text{L.H.S.}$$

$$\left(\frac{16}{81}\right)^{\frac{5}{4}}\div\left(\frac{16}{81}\right)^{\frac{3}{4}}=\left(\frac{16}{81}\right)^{\frac{5}{4}-\frac{3}{4}}$$

Therefore, this follows the rule $x^m \div x^n = x^{m-n}$

EXAMPLE-12. Find the value of each of following-

(i) $\left(\frac{8}{125}\right)^{\frac{4}{3}}\times\left(\frac{8}{125}\right)^{\frac{2}{3}}$ (ii) $\left(\frac{27}{64}\right)^{\frac{7}{3}}\div\left(\frac{27}{64}\right)^{\frac{5}{3}}$

SOLUTION : (i) Rule of rational numbers $a^m \times a^n = a^{m+n}$

$$=\left(\frac{8}{125}\right)^{\frac{4}{3}+\frac{2}{3}}$$

$$=\left(\frac{8}{125}\right)^{\frac{6}{3}\,2}=\left(\frac{8}{125}\right)^{2}$$

$$=\frac{8}{125}\times\frac{8}{125}=\frac{64}{15625}$$



(ii) $\left(\frac{27}{64}\right)^{\frac{7}{3}}\div\left(\frac{27}{64}\right)^{\frac{5}{3}}$

$$=\left(\frac{27}{64}\right)^{\frac{7-5}{3}}=\left(\frac{27}{64}\right)^{\frac{2}{3}}=\left(\frac{3^3}{4^3}\right)^{\frac{2}{3}}$$

$$=\left(\frac{3}{4}\right)^{3\times\frac{2}{3}}=\left(\frac{3}{4}\right)^{2}=\frac{3\times3}{4\times4}=\frac{9}{16}$$

Try This

Find the values of-

(i) $\left(\frac{16}{81}\right)^{\frac{3}{4}} \times \left(\frac{16}{81}\right)^{\frac{5}{4}}$ (ii) $\left(\frac{16}{81}\right)^{\frac{3}{4}+\frac{5}{4}}$ (iii) $\left(\frac{16}{81}\right)^{\frac{5}{4}-\frac{3}{4}}$ (iv) $8^{\frac{2}{3}}$



Which is the Bigger Number?

From 27 and 16, 27 is bigger, but which one is bigger from $\sqrt{16}$ and $\sqrt[3]{27}$?

$\sqrt{16} = \sqrt{\underline{2\times2}\times\underline{2\times2}}$

Means $\sqrt{16} = 2 \times 2$

$= 4$

And $\sqrt[3]{27} = \sqrt[3]{\underline{3\times3\times3}}$

Means $\sqrt[3]{27} = 3$

Therefore $\sqrt{16} > \sqrt[3]{27}$

Similarly,

Is $\sqrt[6]{64}$ bigger or $\sqrt[3]{125}$

$\sqrt[6]{64} = \sqrt[6]{\underline{2\times2\times2\times2\times2\times2}}$

$= 2$

$\sqrt[3]{125} = \sqrt[3]{5\times5\times5}$

$= 5$

Thus, $\sqrt[3]{125} > \sqrt[6]{64}$

Try This



1. Which is the bigger number out of the pair?

   (i) $\sqrt[3]{125}$, $\sqrt{36}$ (ii) $\sqrt{121}$, $\sqrt[3]{729}$

   (iii) $\sqrt[4]{625}$, $\sqrt[5]{1024}$

2. Write in descending order- $\sqrt[3]{125}$, $\sqrt[3]{729}$, $\sqrt[5]{1024}$, $\sqrt{36}$

Surds

An irrational number $\sqrt[p]{a}$ is called a surd, here 'a' is a positive rational number. The sign '$\sqrt{\ }$' is called the sign of the surd (radix). The number p is called the order of the surd and 'a ' is called the radicand (quantity under the radix) of the surd.

$\sqrt{5}, \sqrt{3}, \sqrt{2}$ are irrational numbers, where as $\sqrt[3]{8}$ is a rational number because $\sqrt[3]{8} = 2$. In $\sqrt{5}, \sqrt{3}, \sqrt{2}$ the radicands 5, 3, 2 are positive rational numbers so $\sqrt{5}, \sqrt{3}, \sqrt{2}$ are surds. In $\sqrt[3]{8}$ the radicand 8, is a positive rational number, but $\sqrt[3]{8}$ is not irrational and hence $\sqrt[3]{8}$ is not a surd.

Is $\sqrt{2+\sqrt{3}}$ a surd or not?

Since $\sqrt{3}$ is an irrational number and 2 is a rational number and since the sum of a rational and an irrational number is an irrational numbers. Hence $2+\sqrt{3}$ is also an irrational number. Radicand of the expression $\sqrt{2+\sqrt{3}}$ is irrational therefore this is not a surd.

EXAMPLE-13. Out of these which one are surd?

(i) $\sqrt[3]{5+\sqrt{9}}$ (ii) $\sqrt{\sqrt{3}}$

SOLUTION : (i) $\sqrt[3]{5+\sqrt{9}}$

$= \sqrt[3]{5+3}$

$= \sqrt[3]{8}$

$= \sqrt[3]{2^3}$

$= 2^{3\times\frac{1}{3}}$ $\left[\sqrt[m]{y} = y^{\frac{1}{m}}\right]$

$= 2$

$\because \sqrt[3]{5+\sqrt{9}} = 2$ is a rational number, so it is not a surd.

(ii) $\sqrt{\sqrt{3}}$

$= \sqrt{3^{\frac{1}{2}}}$

$= \left(3^{\frac{1}{2}}\right)^{\frac{1}{2}}$

$= 3^{\frac{1}{4}}$

$= \sqrt[4]{3}$

$\therefore \sqrt{\sqrt{3}} = \sqrt[4]{3}$

So this is a surd.

Try This



1. Identify surd from $\sqrt{3+\sqrt{16}}$, $\sqrt{\sqrt{16}}$, $\sqrt{3+\sqrt{2}}$ and write the reasons for your choice.
2. Write 3 numbers that are irrational but are not surds.

Exercise - 3.2



1. Find the value of each of the following-

(i) $(16)^{\frac{1}{2}}$ (ii) $(243)^{\frac{1}{5}}$ (iii) $(15625)^{\frac{1}{6}}$

2. Simplify each of the following-

(i) $23^{\frac{1}{2}} \times 23^{\frac{3}{2}}$ (ii) $11^{\frac{4}{3}} \times 11^{\frac{5}{3}}$

(iii) $21^{\frac{7}{3}} \div 21^{\frac{1}{3}}$ (iv) $15^{\frac{3}{2}} \div 15^{\frac{5}{2}}$

3. Find the values of-

(i) $\left(\frac{625}{81}\right)^{\frac{4}{3}} \div \left(\frac{625}{81}\right)^{\frac{2}{3}}$ (ii) $\left(\frac{2}{13}\right)^{\frac{4}{3}} \div \left(\frac{2}{13}\right)^{\frac{5}{3}}$

(iii) $3 \times 9^{\frac{1}{2}} \div 9^{\frac{3}{2}}$ (iv) $27^{\frac{2}{3}} \div 27^{\frac{1}{3}} \times 27^{\frac{4}{3}}$

4. Write in an ascending order the surds given in each of the following sets-

(i) $\sqrt{81}, \sqrt[3]{64}, \sqrt[9]{512}$ (ii) $\sqrt[4]{625}, \sqrt{100}, \sqrt[3]{343}$

(iii) $\sqrt[3]{216}, \sqrt[5]{243}, \sqrt{64}$ (iv) $\sqrt[4]{256}, \sqrt[7]{128}, \sqrt[3]{1000}$

5. In the following , find out which are surds and which are not-

(i) $\sqrt{8}$ (ii) $\sqrt[3]{64}$ (iii) $\sqrt{90}$

(iv) $\sqrt{5+\sqrt{2}}$ (v) $\sqrt[5]{2+\sqrt{4}}$

What Have We Learnt



1. When we multiply a number by itself many times, we write that in the exponent form. The number is called the base, for example in 3^6, 6 is the exponent and 3 is the base.

2. The exponent is a way to write very small and very large numbers in a brief and standard form.

3. Multiplicative inverse - The multiplicative inverse of 2^3 is $\frac{1}{2^3}$.

4. Rules of exponents-

$x^m \times x^n = x^{m+n}$ $x^m \div x^n = x^{m-n}$

$(x^m)^n = x^{mn}$ $x^m \times y^m = (xy)^m$

Here x and y are non-zero rational numbers and m and n are rational exponents.

5. Rules for rational exponent are the same as for integer exponents.

